

MAER-Net 2026 PhD Workshop
Chemnitz, 23 September 2026

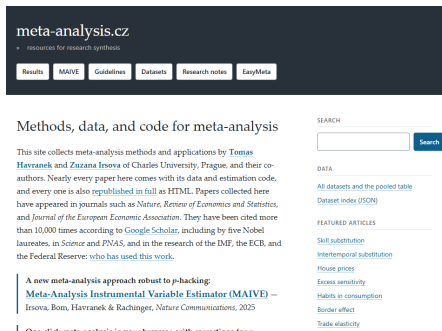
EasyMeta.org and AI in Meta-Analysis



Tomas Havranek

Charles University, Prague

Meta-Research Innovation Center at Stanford



- 71 papers in full, most with data and code
- 46 datasets; one pooled table of about 50,000 estimates
- R replication for 37 papers
- A page written for AI assistants

Model Parameters

Please select the model type and parameters you would like to use.

Model Type

MAVE

Standard Error Treatment

Clustered CR2

Compute Anderson-Rubin Confidence Interval

☒ Yes ☐ No

Note that selecting WLS in Model Type allows you to run classical PET, PEESE, PET-PEESE, and EK.

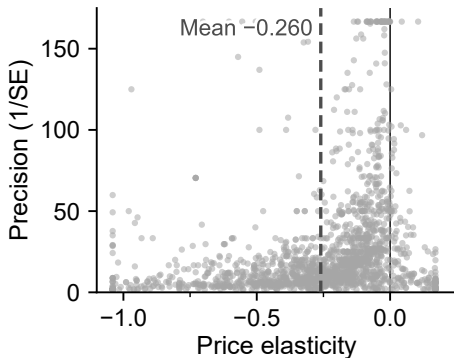
Advanced Options [➤](#)

Run Model

- Upload effect, SE, sample size, study ID
- PET-PEESE, EK, MAIVE, WAIVE, RTMA, RDT
- Clustering, winsorizing, weights, Anderson-Rubin intervals
- A reproducibility package to download

Built by Petr Cala. Also runs through an API.

Demo data: electricity demand



Short-run price
elasticities: 1,647
estimates from 226
studies, winsorized 2.5%

Kudela, Havranek, Irsova,
Kudelova & Sikl (2026),
meta-analysis.cz/electricity

Try it

1. Download from meta-analysis.cz/chemnitz
2. Upload at easymeta.org
3. Model WLS, winsorize 2.5%, Run

Pasted into Claude Code (excerpt)

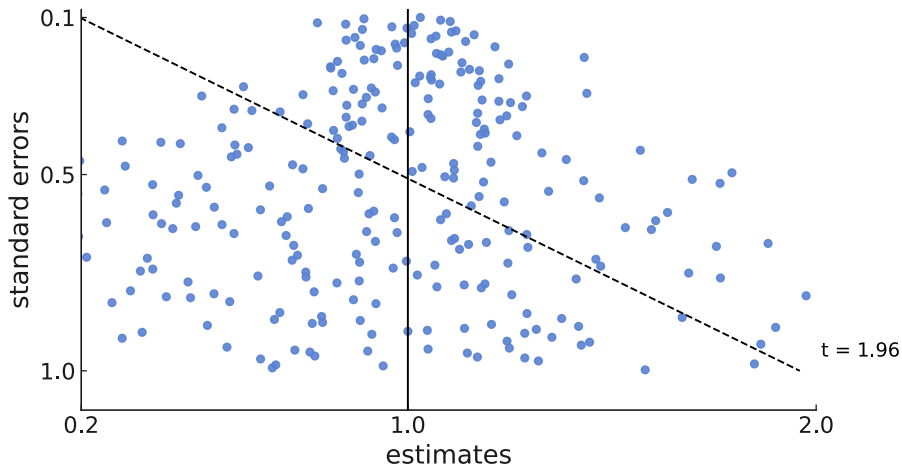
```
Use the EasyMeta API, not your own estimation. [...]
Send every row unchanged to
POST https://api.maive.eu/v1/run-model [...]
{"recipe":"PET-PEESE","parameters":{"winsorize":2.5},
 "data":[...] } and the same with "recipe":"EK". [...]
Never report a number the API did not return.
```

	App	API, through Claude
PET-PEESE	-0.127	-0.127104860252972
EK	-0.105	-0.10525106160908

Matching digits show only that the code ran as asked.

Full prompt: meta-analysis.cz/chemnitz

No selection, no bias

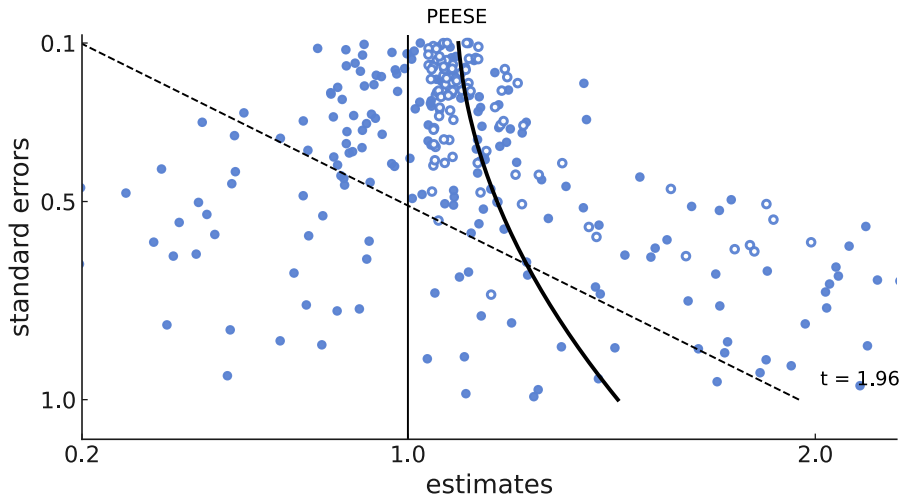


10 / 44

12/44

13/44

Aggressive p -hacking: PEESE in trouble



Could this happen? Meta of education premium

True model:

The diagram shows the equation $Earnings_j = \gamma Education_j + \delta Ability_j + v_j$. A curved arrow points from the $Education_j$ term to the $Ability_j$ term, and another curved arrow points from the $Ability_j$ term back to the $Education_j$ term, indicating a causal relationship between the two variables.

$$Earnings_j = \gamma Education_j + \delta Ability_j + v_j,$$

Ability **not observed**.

Primary studies:

- 1 ignore ability $\rightarrow \hat{\gamma}$ too large, $SE(\hat{\gamma})$ too small.
- 2 include a proxy $\rightarrow \hat{\gamma}$ smaller, $SE(\hat{\gamma})$ larger.
- 3 quasi-experiment $\rightarrow \hat{\gamma}$ even smaller, $SE(\hat{\gamma})$ even larger.

Could this happen? Meta of education premium

Omitted variable:

$$Earnings_j = \gamma Education_j + w_j,$$

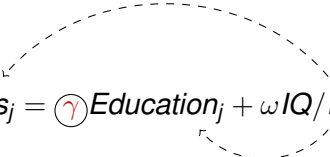
Ability **not observed**.

Primary studies:

- 1 ignore ability $\rightarrow \hat{\gamma}$ too large, $SE(\hat{\gamma})$ too small.
- 2 include a proxy $\rightarrow \hat{\gamma}$ smaller, $SE(\hat{\gamma})$ larger.
- 3 quasi-experiment $\rightarrow \hat{\gamma}$ even smaller, $SE(\hat{\gamma})$ even larger.

Could this happen? Meta of education premium

Proxy:


$$Earnings_j = \gamma Education_j + \omega IQ/PGS_j + x_j,$$

Ability **not observed**.

Primary studies:

- 1 ignore ability $\rightarrow \hat{\gamma}$ too large, $SE(\hat{\gamma})$ too small.
- 2 include a proxy $\rightarrow \hat{\gamma}$ smaller, $SE(\hat{\gamma})$ larger.
- 3 quasi-experiment $\rightarrow \hat{\gamma}$ even smaller, $SE(\hat{\gamma})$ even larger.

Could this happen? Meta of education premium

Instrument:

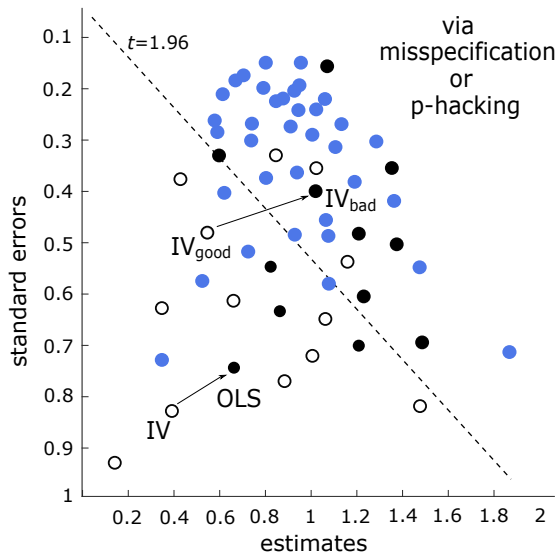
$$Earnings_j = \gamma Education_j + z_j, \quad \text{Policy reform}$$

Ability **not observed**.

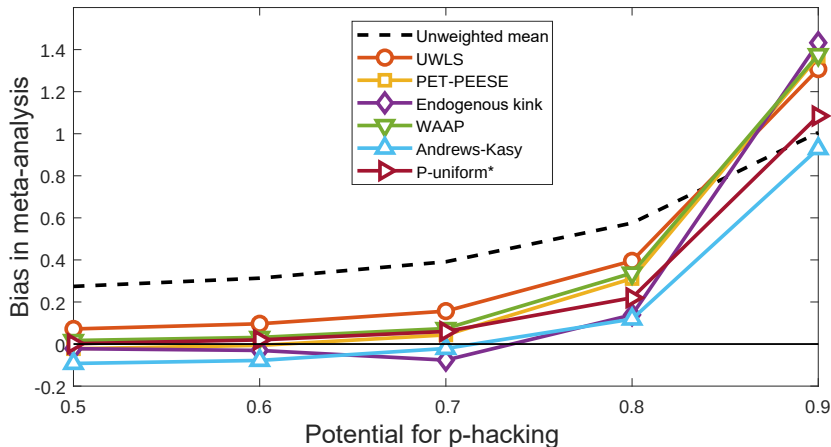
Primary studies:

- 1 ignore ability $\rightarrow \hat{\gamma}$ too large, $SE(\hat{\gamma})$ too small.
- 2 include a proxy $\rightarrow \hat{\gamma}$ smaller, $SE(\hat{\gamma})$ larger.
- 3 quasi-experiment $\rightarrow \hat{\gamma}$ even smaller, $SE(\hat{\gamma})$ even larger.

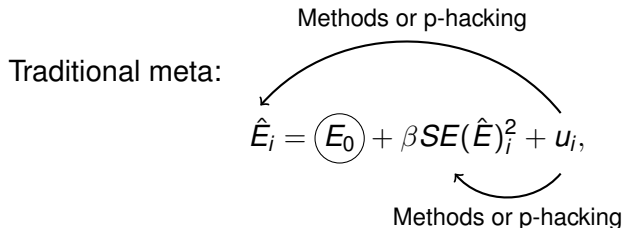
Some estimates spuriously large & precise



All meta estimators biased upwards



Why? Key meta assumption broken



$\text{corr}(SE, u) \neq 0 \Rightarrow \hat{\beta} \text{ and } \hat{E}_0 \text{ biased.}$

Natural solution: 1/N instrumenting $SE(\hat{E}_i)^2 \rightarrow \text{MAIVE.}$

Meta-analysis instrumental variable estimator

MAIVE intuition:

Definition of SE

$$\hat{E}_i = \underbrace{(E_0)}_{\text{Definition of SE}} + \beta SE(\hat{E})_i^2 + u_i. \quad 1/N_i$$

- Can **add controls**.
- Plug MAIVE-adjusted SEs into other estimators.
- In practice it's better to use logs in the first stage.

Meta-analysis instrumental variable estimator

MAIVE first stage:

$$SE(\hat{E})_i^2 = \alpha_0 + \alpha_1 (1/N_i) + \pi_i.$$

hacking or misspecifications



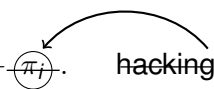
- Can **add controls**.
- Plug MAIVE-adjusted SEs into other estimators.
- In practice it's better to use logs in the first stage.

Meta-analysis instrumental variable estimator

MAIVE adjustment:

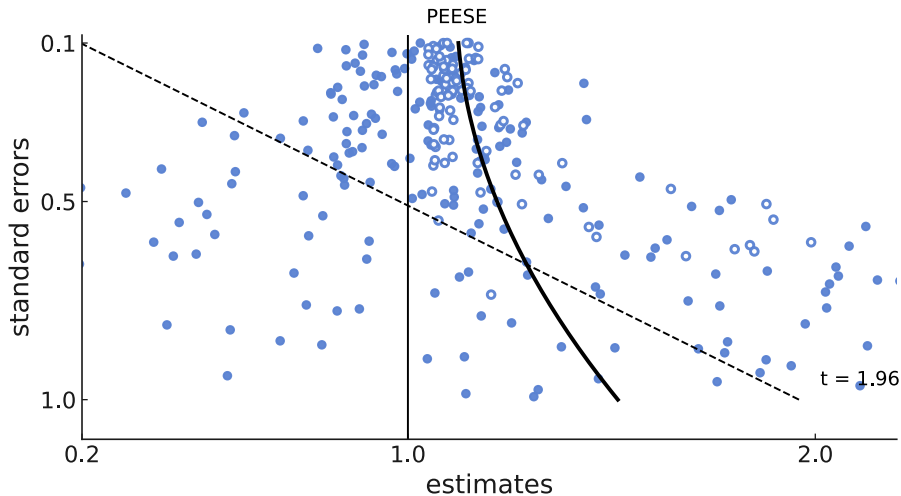
$$SE(\hat{E})_{adj,i}^2 = \hat{\alpha}_0 + \hat{\alpha}_1 (1/N_i) + \cancel{\pi_i}.$$

hacking or misspecifications

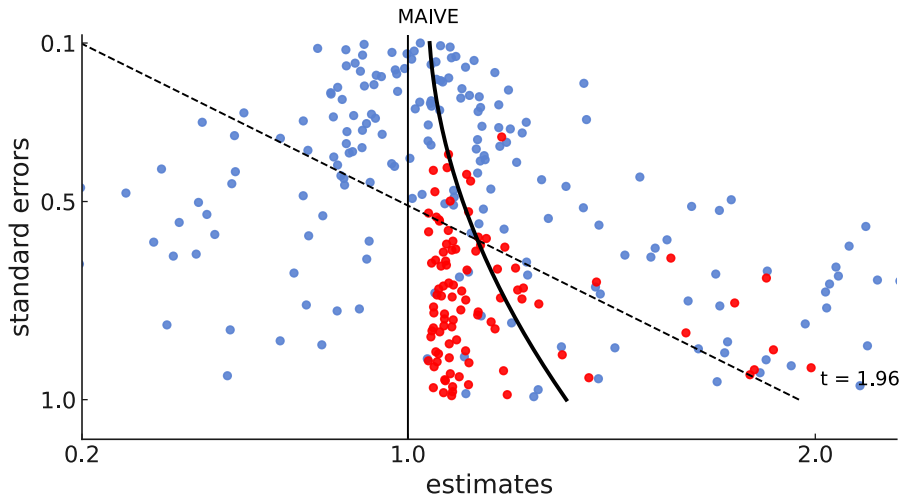


- Can **add controls**.
- Plug MAIVE-adjusted SEs into other estimators.
- In practice it's better to use logs in the first stage.

Aggressive p -hacking: PEESE in trouble



Meta-Analysis Instrumental Variable Estimator



MAIVE published last year

nature communications

Explore content ▾ About the journal ▾ Publish with us ▾

[nature](#) > [nature communications](#) > [articles](#) > article

Article | [Open access](#) | Published: 26 September 2025

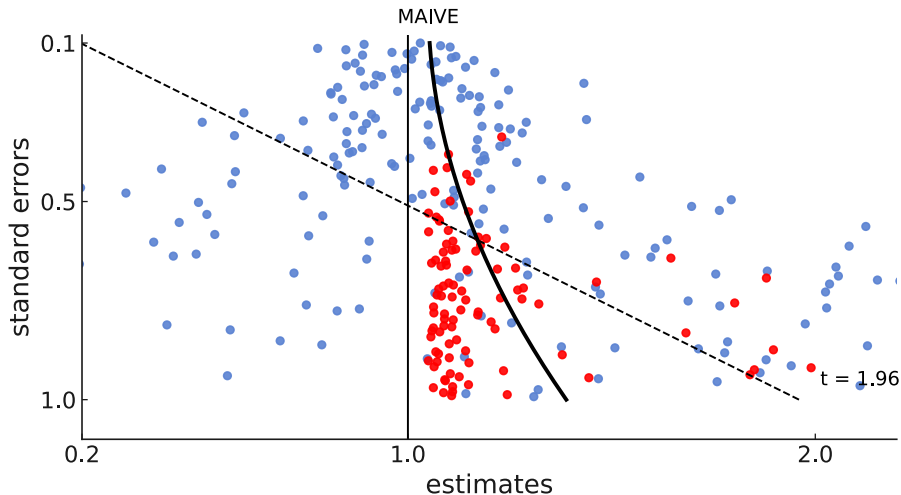
Spurious precision in meta-analysis of observational research

[Zuzana Irsova](#) ✉, [Pedro R. D. Bom](#), [Tomas Havranek](#) & [Heiko Rächinger](#)

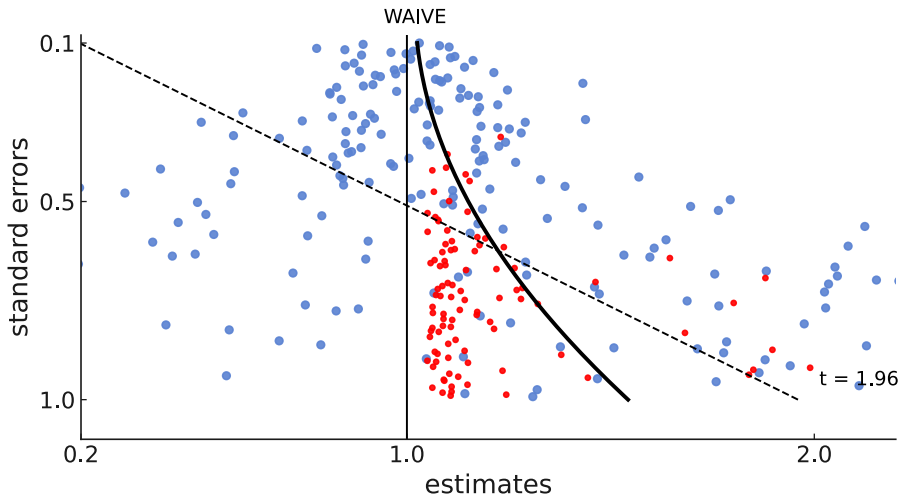
[Nature Communications](#) **16**, Article number: 8454 (2025) | [Cite this article](#)

1036 Accesses | **15** Altmetric | [Metrics](#)

Meta-Analysis Instrumental Variable Estimator

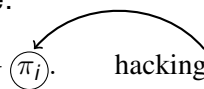


Weighted Adjustment Instrumental Variable Estimator



WAIVE first stage – same as MAIVE

Residual from MAIVE first stage:

$$SE(\hat{E})_i^2 = \alpha_0 + \alpha_1 (1/N_i) + \pi_i.$$


hacking if negative?

- Negative $\pi_i \rightarrow$ reported **SE too small relative to N** $\rightarrow$ possibly p -hacking.
- Analogy: profit shifting in accounting (regress profits on sales, look at residuals).

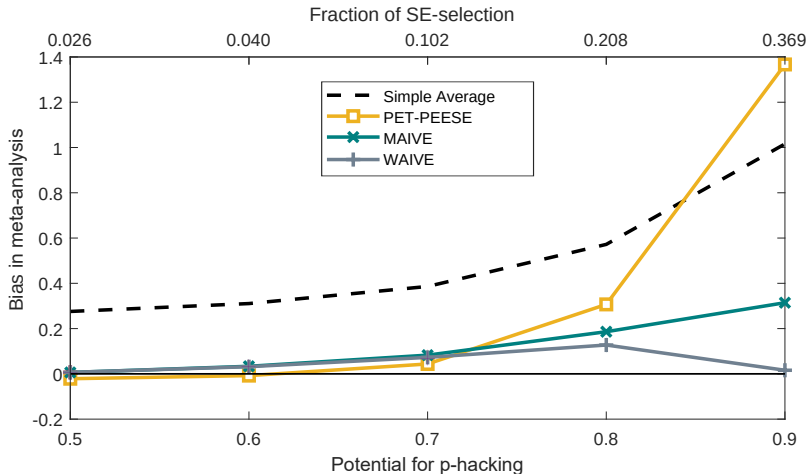
WAIVE weight for the second stage

Exponential tilt:

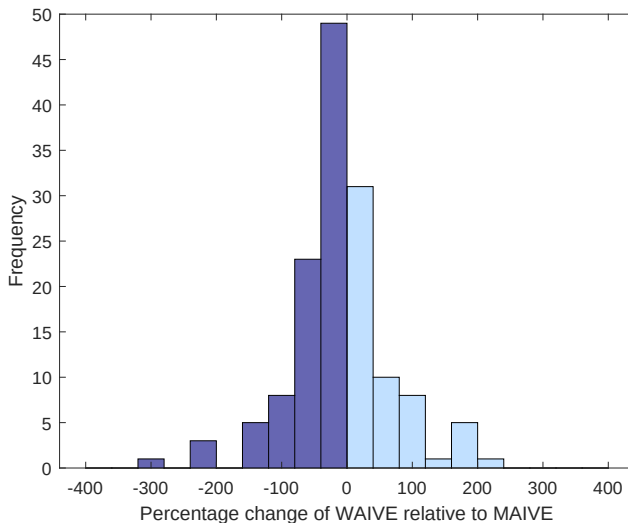
$$\omega_i = \exp[-\max(0, -\pi_i)]$$

- **Penalize only negative residuals** ($\pi_i < 0$) – downweight spuriously precise estimates.
- Weight $\in (0, 1]$, smooth and proportional decay.
- Then run PEESE of $\hat{E}_i$ on $SE(\hat{E})_{adj,i}^2$ with weights ω_i .

WAIVE alleviates the bias



WAIVE reduces MAIVE in 2/3 of meta-analyses



Practical issues

WAIVE still just an experimental concept.

Main consideration

- Can methods or p -hacking affect SE ?
- Yes → MAIVE and WAIVE help ([EasyMeta.org](https://easymeta.org)).
- No → MAIVE and WAIVE don't hurt much (but wider CI).

Project Website

meta-analysis.cz/maive

Papers using MAIVE



Irsova Z., P. Bom, T. Havranek, & H. Rachinger (2025):
Spurious Precision in Meta-Analysis of Observational
Research.

Nature Communications 16: 8454.



Opatrny M., T. Havranek, Z. Irsova, & M. Scasny (2026):
Publication Bias and Model Uncertainty in Measuring the
Effect of Class Size on Achievement.

Journal of Labor Economics, forthcoming.



Cala P., T. Havranek, Z. Irsova, M. Luskova, J. Matousek
& J. Novak (2026): Financial Incentives and Performance:
A Meta-Analysis of Experiments in Economics.

Journal of Political Economy Microeconomics, forthcoming.

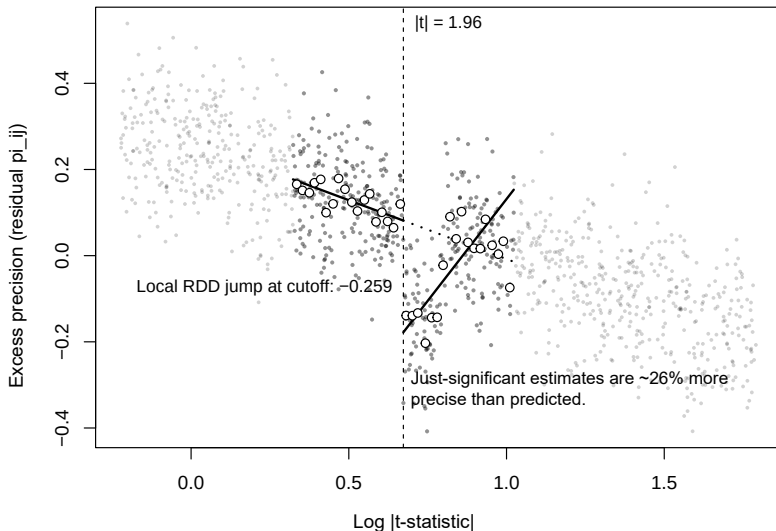
Diagnostic: excess precision

Residual from meta-regression:

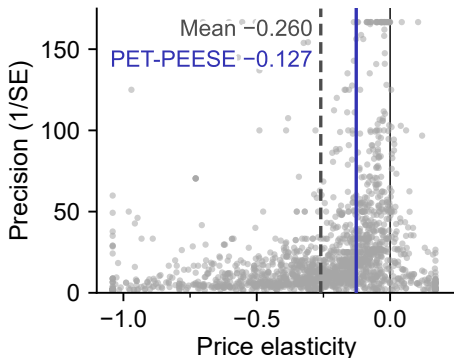
$$\log \text{SE}(\hat{E}_i) = \alpha_0 + \alpha_1 \log N_i + \underbrace{\pi_i}_{<0: \text{excess precision}}$$

- Negative $\pi_i \rightarrow$ reported **SE too small relative to N** $\rightarrow$ may indicate p -hacking.
- Interpretation: $\pi_i = -0.25$ means SE is roughly one quarter smaller than expected.

RDT: residual discontinuity test (simulated)



Your turn: where does MAIVE land?



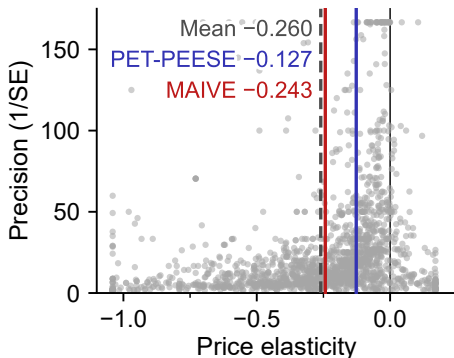
Mean -0.260
PET-PEESE -0.127

**MAIVE: closer to zero,
or further?**

Try it

Start again at easymeta.org,
not Back. Model MAIVE;
Advanced Options: log first
stage Yes, winsorize 2.5%;
Run

MAIVE: -0.24



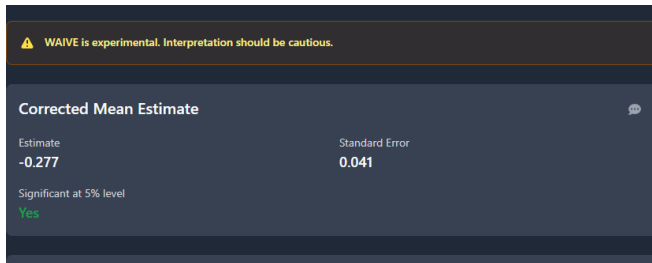
PET-PEESE -0.127
equal weights -0.226
MAIVE -0.243

Log first stage, $F = 25$;
SE 0.038, clustered by study

42 estimates report SEs
far below what their
sample size predicts.
They carry 37% of
PET-PEESE's weight and
are small in absolute
value.

Levels first stage: $F = 6.9$, and
the app warns.

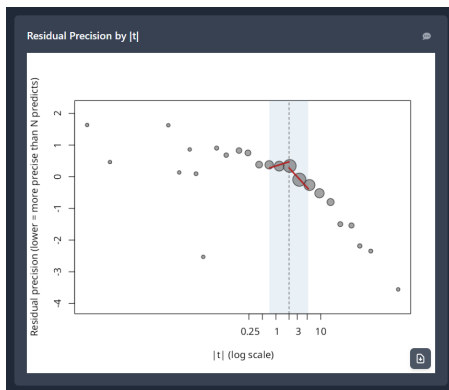
WAIVE: -0.28



MAIVE -0.24 ; WAIVE downweights what is left: -0.28 .

Experimental. Fixed in September: the package had applied WAIVE's weights the wrong way round (MAIVE issue #30).

RDT in EasyMeta



Jump at $|t| = 1.96$: -0.19 ($p = 0.13$). A single literature gives this test little power.

App only: the API does not offer RDT.

MAER-Net guidelines, updated for AI

Cook et al. (2026a, 2026b), *Journal of Economic Surveys*. Thirteen authors, among them Bartos, Stanley, Gechert and Prante.

Human leadership

People direct every stage.

Accountability

AI is never a co-author.

Auditing

Humans recheck a random 10% of what AI screens and codes.

Disclosure

Any AI use that shapes the results is disclosed.

Recommendations; deviations should be disclosed. Free at meta-analysis.cz/guidelines.

New items in the reporting checklist

Stage	New in 2026
Search	AI use disclosed; AI search logs shared
Screening	random audit of 10% (at least 100 records); false-negative rate
Coding	AI may be a coder; humans recode 10% (at least 20 studies); kappa
Data	a dummy for human-reviewed estimates; share coded by AI
Figures	“Figure produced by AI.” if not validated
Replication	tool, version, dates, prompts, settings

Cook et al. (2026b) replaces the 2020 guidelines for every meta-analysis, with or without AI.

Does debate help? A pre-registered test

- 55 published or accepted meta-analyses in economics: 27 ours, 28 from JoES since 2022
- Three AI reports per paper: one prompt to one model, mad-research, paper-workshop
- Same template and length, tool names removed, random order
- Authors of 44 of them rank the reports by usefulness

Pre-registered on OSF on 22 June 2026, before any report was written.

The single pass: this prompt, one call

79-word base prompt, given to all three arms

You are refereeing a meta-analysis submitted to an economics journal. Read the attached paper and write a referee report whose goal is to help the authors improve it. Be specific and constructive, and point to concrete parts of the paper.

Structure the report as:

1. Summary — one short paragraph: what the paper does and its main claim.
2. Major comments — the most important issues, most serious first.
3. Minor comments.
4. Overall assessment.

Keep it concise (about 1,000 words).

Verbatim from the replication package, doi 10.5281/zenodo.21273528.

Practical takeaway

Publication bias and p -hacking

Report several corrections side by side.

Reported precision

If studies report sample size, **MAIVE** instruments the standard error with it. **WAIVE** (experimental) goes further and downweights estimates still too precise for their sample size.

AI

Recheck a random 10% of what AI codes (at least 20 studies) and keep the prompts. For feedback, start with one good prompt.

Try it

easymeta.org · meta-analysis.cz/chemnitz

Selected recent papers



Irsova Z., P. Bom, T. Havranek, & H. Rachinger (2025):
Spurious Precision in Meta-Analysis of Observational
Research.

Nature Communications 16: 8454.



Havranek T., & Z. Irsova (2026):

Does Multi-Agent Debate Improve AI Feedback on Research Papers?

arXiv:2607.14713.



Cala P., T. Havranek, Z. Irsova, M. Luskova, J. Matousek & J. Novak (2026): Financial Incentives and Performance: A Meta-Analysis of Experiments in Economics.

Journal of Political Economy Microeconomics, forthcoming.

Claude Opus 5, Claude Fable 5.1 and GPT-6 Astra were used in preparing this demo and presentation.

Backup: upload and column mapping

MAIVE UI

My Runs ⚙️ ⓘ 🏠

Map Your Columns

Tell MAIVE which columns contain your effect sizes, standard errors, and sample sizes.
Each column can only be mapped once. The Sample size field is required for every model except RTMA; without it, only RTMA will be available. The Study ID field is optional: leave it blank if you don't have study-level clustering.

We've pre-selected columns where the headers looked familiar. Feel free to adjust before continuing.

Effect size*

Standard error*

effect

se

Sample size

Study ID (optional)

n_obs

study_id

Data preview

Values shown here update automatically as you change the column mapping.

EFFECT SIZE (EFFECT)	STANDARD ERROR (SE)	SAMPLE SIZE (N_OBS)	STUDY ID (OPTIONAL) (STUDY_ID)
-0.04	0.023	28	5
-0.07	0.037	28	5
-0.06	0.022	28	5
-0.08	0.040	28	5
-0.09	0.028	28	5

Total rows: 1647 (showing first 5 rows)

Use a subsample filter?

Would you like to include only a subset of your data in the analysis?

All four columns map automatically.

Backup: WLS, PET-PEESE, winsorize 2.5%

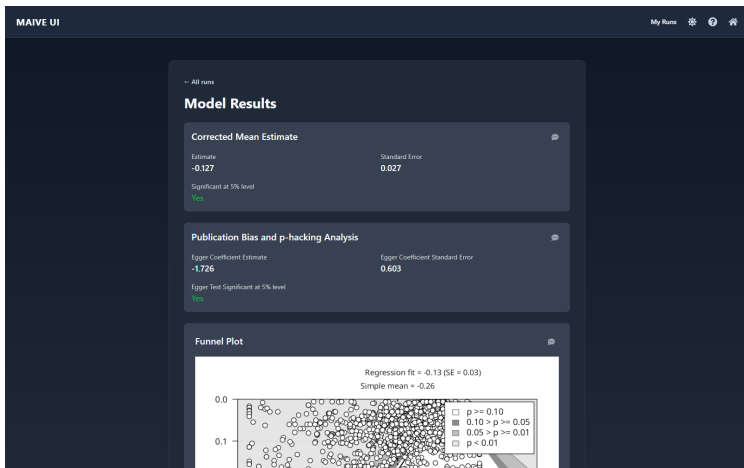
The screenshot shows the 'Model Parameters' configuration interface of the MAIVE UI. The interface is dark-themed with white and blue text. At the top, there's a 'Back to Validation' link. The main section is titled 'Model Parameters' and contains several configuration options:

- Model Type:** A dropdown menu set to 'WLS'.
- Standard Error Treatment:** A dropdown menu set to 'Clustered CR2'.
- Advanced Options:** A section with a downward arrow, containing:
 - Method:** A dropdown menu set to 'PET-PEESE'.
 - Weighting:** A dropdown menu set to 'Standard Weights'.
 - Winsorization (%):** A slider control set to 2.5%.
 - Fixed-Intercept Multilevel:** Radio buttons for 'Yes' and 'No', with 'No' selected.

A blue 'Run Model' button is at the bottom. A blue informational box states: 'Without instrumenting, this run is not MAIVE. Please still cite Iversen et al., Nature Communications (2020) as the app used using the Cite button in the footer if you use this tool in your research.'

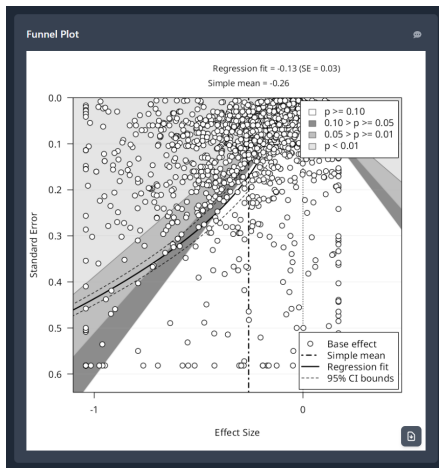
Settings for the first run.

Backup: PET-PEESE

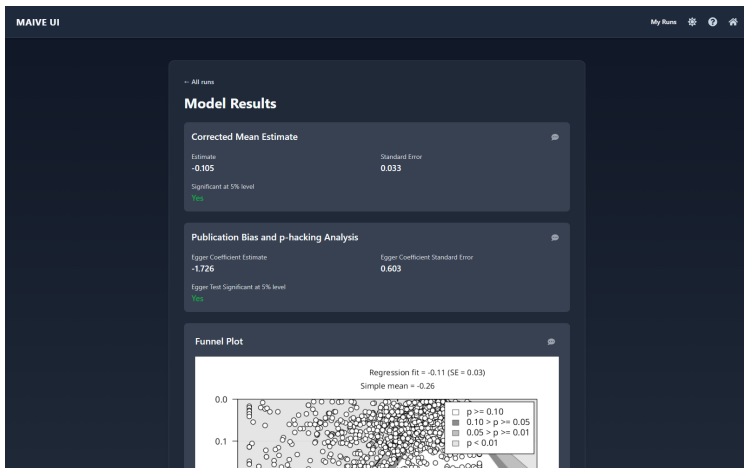


PET-PEESE picks PEESE: -0.127 .

Backup: funnel plot in the app



Same estimates as on the demo slide; the app puts SE on the vertical axis, 0 at the top.



EK finds no kink, so it equals PET: -0.105 .

Backup: what the API returned

```
POST https://api.maive.eu/v1/run-model
1,647 rows, winsorize 2.5
```

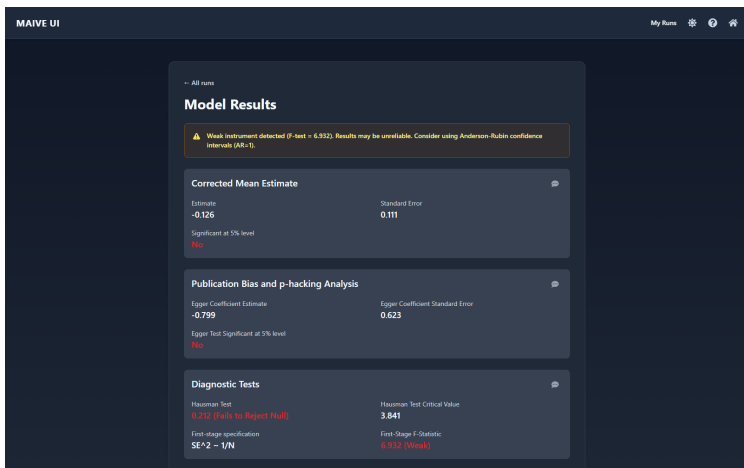
```
recipe PET-PEESE      HTTP 200
  effectEstimate      -0.127104860252972
  standardError        0.0269191742163644      (PEESE selected)
```

```
recipe EK             HTTP 200
  effectEstimate      -0.10525106160908
  standardError        0.0327033930987689
```

```
MAIVE, log first stage HTTP 200
  effectEstimate      -0.24255281691904
  standardError        0.037775202532599
  firstStageFStatistic 25.03
```

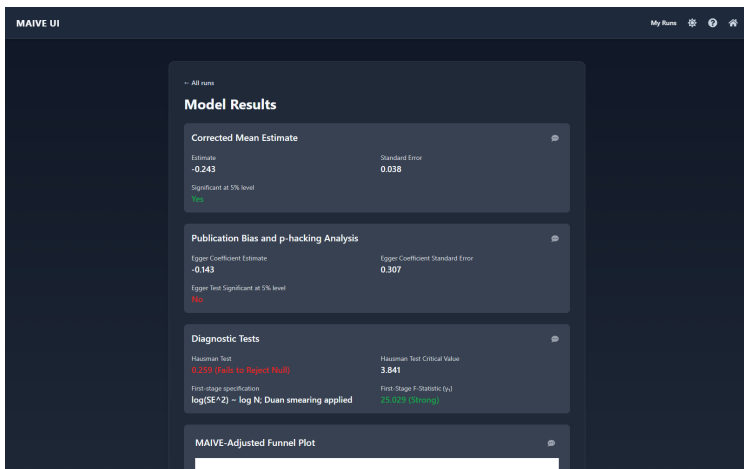
```
WAIVE, same settings  HTTP 200
  effectEstimate      -0.277390923733746
```

Backup: MAIVE, levels first stage



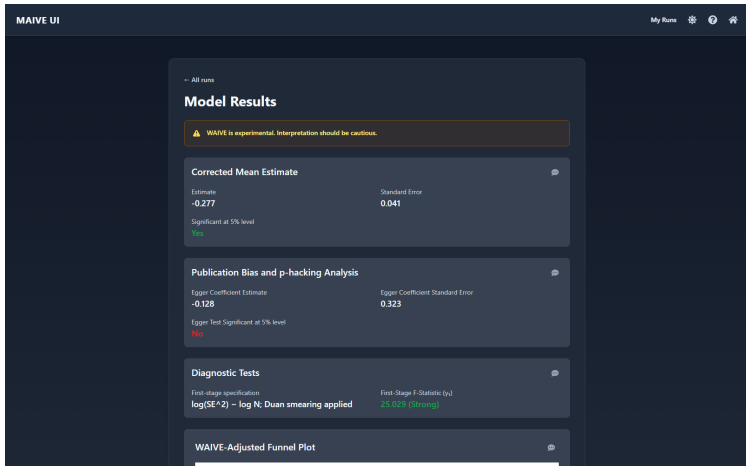
Levels first stage: MAIVE -0.126 , $F = 6.9$, and the app warns.

Backup: MAIVE, log first stage



Log first stage: $F = 25$, MAIVE -0.24 .

Backup: WAIVE



WAIVE -0.28, fixed (MAIVE issue #30).

Backup: RDT

RDT Results

Jump in residual precision at $|t| = 1.96$

-0.192 [-0.440, 0.057]

SE 0.122, $p = 0.126$; negative = estimates just past the threshold are more precise than their sample sizes predict

Estimation window

$|t|$ from 0.72 to 5.31 (bandwidth 1.00 log points)

376 below and 688 above the cutoff inside it, 193 studies; 1647 rows used, 0 dropped

Smallest jump this dataset could detect (80% power)

0.352

a jump of about 0.10 corresponds to standard errors roughly 10% smaller than sample size predicts; below the detectable size, a null result says nothing.

Jump -0.19 ($p = 0.13$): no clear jump at $|t| = 1.96$.